LESSON 6.13

Prime Factorization

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write a number as a product of its prime factors using a factor tree.

WHY THIS MATTERS The space station received 60 supply crates. Mission Control needs to break this quantity into its prime components so the sorting robots can distribute them into equally sized pods. Help the crew find the prime factorization of 60!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Prime Factorization *Spanish: Factorización prima*	Writing a whole number as a product of only prime numbers.	36 = 2 × 2 × 3 × 3 = $2^2 \times 3^2$
Prime number *Spanish: Número primo*	A number bigger than 1 that you can only divide by 1 and itself.	7 has only two factors: 1 × 7. So 7 is prime.
Composite number *Spanish: Número compuesto*	A number bigger than 1 that you can divide by more than just 1 and itself.	12 = 1 × 12, 2 × 6, 3 × 4 — six factors, so 12 is composite
Factor *Spanish: Factor*	A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because 12 ÷ 3 = 4.	Factors of 12: 1, 2, 3, 4, 6, 12 — each divides 12 evenly.
Factor tree *Spanish: Árbol de factores*	A picture that splits a number into its prime numbers, step by step.	24 → 4 × 6 → (2 × 2) × (2 × 3) → 2 × 2 × 2 × 3
Exponent *Spanish: Exponente*	A small number that tells how many times to multiply a number by itself.	2^3 means 2 × 2 × 2 = 8

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want the prime factorization of 60. I start by splitting it: 60 = 6 × 10.

1. 6 is not prime, so I break it: 6 = 2 × 3. Both 2 and 3 are prime, so I stop those branches.
2. 10 is not prime, so I break it: 10 = 2 × 5. Both 2 and 5 are prime, so I stop.
3. Now every factor is prime: 60 = 2 × 2 × 3 × 5.
4. Using exponents, I write it as $2^2 \times 3 \times 5$. I check: 2 × 2 × 3 × 5 = 60. Correct.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's factor 12 together. What is one way to split 12 into two factors? (You might say 3 × 4.)

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I know _____ is prime because its only factors are _____ and _____. I know _____ is composite because it has factors like _____.

Word bank Prime Factorization | Prime number | Composite number | Factor | Factor tree | Exponent | prime | composite | factors | divisible | two factors | more than two

WATCH OUT!

A common mistake

A common mistake in Prime Factorization is stopping the factor tree too soon

LESSON 6.13

Prime Factorization

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each expression: is it a COMPLETE prime factorization, or does it still have composite factors?

Groups: Complete Prime Factorization | NOT Complete (Has Composite Factors)

_____ 2 × 2 × 3 (for 12)

_____ 2 × 3 × 5 (for 30)

_____ 2 × 2 × 2 × 3 (for 24)

_____ 4 × 3 (for 12)

_____ 6 × 5 (for 30)

_____ 8 × 3 (for 24)

STEPS TO TRY

1. 'Complete' means every single factor shown is prime. Even one composite factor makes the whole expression NOT complete.
2. Scan each expression for numbers like 4, 6, or 8 — those are composite, so any expression containing them isn't complete.
3. Take '4 × 3 (for 12)': 4 = 2 × 2, so 4 is composite. That card goes in NOT Complete. Check each other card's factors the same way.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which of the following is a prime number?

(A) 9

(B) 15

(C) 17

(D) 21

STEPS TO TRY

1. Re-read what 'prime' means: exactly two factors, 1 and itself. Any extra factor makes a number composite.
2. Test each choice by dividing by small primes like 2, 3, 5, and 7. If anything divides in evenly, that number is composite.
3. Start with 9: since 3 × 3 = 9, it has a factor besides 1 and 9, so 9 is composite. Cross it off and test the next choice the same way.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the prime factorization of 30?

(A) 2 × 3 × 5

(B) 2 × 15

(C) 3 × 10

(D) 5 × 6

STEPS TO TRY

1. A prime factorization uses ONLY prime numbers. Look at each choice: are ALL of its factors prime?
2. Break 30 down step by step, and keep splitting any factor that is still composite until every factor is prime.
3. Start with 30 = 2 × 15. Is 15 prime? No — so split 15 into two primes and collect everything you found.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the prime factorization of 18?

(A) 2 × 3 × 3

(B) 2 × 9

(C) 3 × 6

(D) 6 × 3

STEPS TO TRY

1. Check every factor in each choice — numbers like 9 and 6 are composite, so they can't appear in a prime factorization.
2. Build a factor tree for 18: split it into two factors, then keep splitting anything composite until only primes remain.
3. Begin with 18 = 2 × 9. The 9 isn't prime, so break 9 into two equal prime factors and list all the primes you end with.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide whether a number is prime or composite? What strategy did you use?

SENTENCE STARTER I know ___ is prime because its only factors are ___ and ___. I know ___ is composite because it has factors like ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help